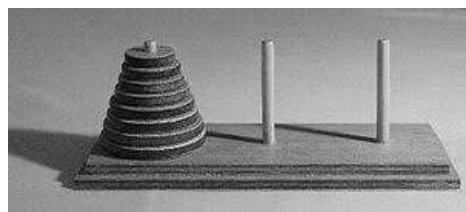


Option 1 of Project: Tower of Hanoi – Recursion and Algorithm Design

The Towers of Hanoi is a classic mathematical **puzzle** [here](https://www.amazon.com/Designs-Teasers-Challenges-Mastermind-Learning/dp/B00QIH5VWQ/ref=sr_1_2_sspa?crid=20AWUKEPTV5YL&dib=eyJ2ljojMSJ9.LBb_d9Qo0fnvFgF-pbvstf9UrFiA4xQlbm2HdYt_VCRsyaHUINozyP81agXDimwQIJG1Zorx3EDvUdl04F-goNYkQT37PLv8_mhXrQANZQNqR_2Jn48huwvVKFC8mi1q2CDzjpntybdnuDpT0WgZ0rpFzvv2rvKEKSbfQ5NeUhiFM4ORfH7Invg8QFrf3VnqYbucG2up35WTXXieQojYIy2h3jHNBCGYL9VKVXAqb3mO_Ajo1wt3XR6W-aRE17isiltj0oDg99ZFFVJkLyliNmHRAOOg7ROw958Vx9B9zs8s_oNq3FzBavb_nXBNH4iFsv47u7s46pcDv_-Qjwhxi8&dib_tag=se&keywords=towers+of+hanoi&qid=1708374272&srefix=towers+of+hanoi%2Caps%2C139&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1)) that consists of three pegs and a number of disks of different sizes which can slide onto any peg. The puzzle starts with the disks neatly stacked in ascending order of size on one peg, the smallest at the top, thus making a conical shape.



You can learn more about the puzzle [here](https://youtu.be/_bWLwz_Pqul?si=JXNos5zkHShnWych)). Before doing this project I urge you to play the puzzle at home either with the real wooden pieces, or with a paper version by cutting up circular disks (preferably using construction, or cardboard paper). You don't necessarily need the pegs.

1. Written Report (PDF or DOCX)

Must include:

- Explanation of the Tower of Hanoi problem and rules
- Derivation of the recursive formula $T(n) = 2T(n-1) + 1$, and closed form/explicit formula $T(n) = 2^n + 1$. Be sure to explain the recursive formula (in your own words), and be sure to derive the explicit formula using mathematical induction.
- Pseudocode of the recursive algorithm. Diagram of the way the pseudocode works just as in Participation Activity 6.9.1, or 6.9.2. You're welcome to modify the representation in ways that makes sense to you. Finally, be sure to provide a proof of the validity of the recursive algorithm using the similar methodology as in the Section 6.10 of zybook.
- Full worked-out example ($n = 4$) in using description of the steps (e.g. Move disk 1 to peg 2, move disk 2 to peg 3, etc...)

- Time complexity analysis and reflection on the efficiency of the algorithm. Compare the algorithm to the recursive algorithm for computing the fibonacci sequence. The latter, for example, has a lot of redundant calculations. Is the same true for your algorithm for the towers of hanoi? Why or why not?
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2. Code Implementation

- Recursive solution in Python, JavaScript, or another language
 - Accepts input (number of disks)
 - Outputs move list (e.g., “Move disk 1 from A to C”)
 - Code should be readable and commented
 - (Optional) Add visualization (ASCII art, text towers, or animation)
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3. Video Presentation (5–8 minutes)

- Explain the rules and objective of the puzzle
 - Describe the recursive approach with diagram or tree
 - Walk through your algorithm with a small example
 - Demonstrate the code working and output
 - Reflect on the mathematical insight or difficulty
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Rubric (100 Points Total)

Category	Points	Criteria
Concept Explanation	20	Clear explanation of the Tower of Hanoi rules and the recursive logic. Includes derivation of the recurrence formula and closed-form solution. The proofs using induction of the closed formula.
Algorithm & Pseudocode	15	Accurate and complete pseudocode. Describes how the recursive solution works in steps. Discussion of Efficiency (either on paper, or on video). Proof using induction of the validity of the pseudocode.
Code Implementation	15	Correct recursive implementation. Accepts input and outputs correct move sequence. Code is commented and readable.
Worked Example	10	A complete trace of the solution for $n = 3$ or 4 , shown step-by-step, and implemented in code.
Complexity Analysis	10	Explains why time complexity is exponential. Describes recursive depth and behavior.

Category	Points	Criteria
Written Report Quality	10	Report is clear, complete, well-organized, and includes all required sections.
Video Presentation	15	Engaging, clear, and within time. Shows algorithm, code, and example. Uses visuals effectively.
Creativity and Depth	5	Shows extra effort through clear visuals, animations, or thoughtful reflection.